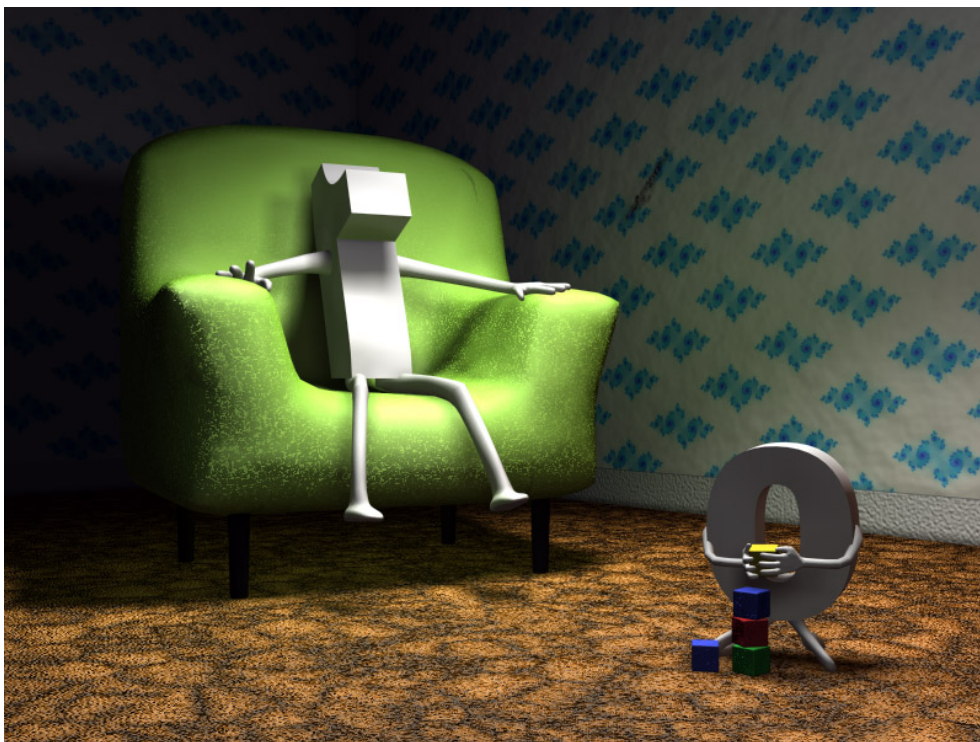


Learning how to prove Workshop Week 1

Integers



**Induction. Division Algorithm. Gcd. Primes and coprimes
integers. Euclidean Algorithm.**

1. Prove each statement by induction on n .

- (a) $\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \dots + \frac{1}{\sqrt{n}} \geq \sqrt{n}$, for all $n \geq 1$.
 (b) If $\gcd(a, b) = 1$, prove that $\gcd(a, b^n) = 1$, for all $n \geq 1$.

Solution: (a) We will use the following two properties of the radicals:

$$\begin{aligned}\sqrt{ab} &= \sqrt{a}\sqrt{b}, \\ a \geq b &\Rightarrow \sqrt{a} \geq \sqrt{b}\end{aligned}$$

Let $n \geq 1$ and define

$$p_n : \quad \frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \dots + \frac{1}{\sqrt{n}} \geq \sqrt{n}$$

It is clear that p_1 is true, since

$$\frac{1}{\sqrt{1}} = 1 \geq 1 = \sqrt{1}.$$

Next, take $k \geq 1$ and assume that p_k is true. To prove that p_{k+1} holds, we first apply the hypothesis induction

$$\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \dots + \frac{1}{\sqrt{k}} + \frac{1}{\sqrt{k+1}} \geq \sqrt{k} + \frac{1}{\sqrt{k+1}}. \quad (1)$$

Let us do some calculations:

$$\sqrt{k} + \frac{1}{\sqrt{k+1}} = \frac{\sqrt{k}\sqrt{k+1} + 1}{\sqrt{k+1}} = \frac{\sqrt{k(k+1)} + 1}{\sqrt{k+1}} = \frac{\sqrt{k^2 + k} + 1}{\sqrt{k+1}} \geq \frac{k+1}{\sqrt{k+1}}, \quad (2)$$

since

$$k^2 + k \geq k^2 \Rightarrow \sqrt{k^2 + k} \geq \sqrt{k^2} = k.$$

From (1) and (2) we obtain that

$$\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \dots + \frac{1}{\sqrt{k}} + \frac{1}{\sqrt{k+1}} \geq \frac{k+1}{\sqrt{k+1}} = \sqrt{k+1},$$

which shows that p_{k+1} is true. The Principle of Mathematical Induction finishes the proof.

(b) Let $n \geq 1$ and define the statement $p_n : \gcd(a, b^n) = 1$. Observe that p_1 is true since $\gcd(a, b) = 1$ by hypothesis. Assume that p_k is true for $k \geq 1$, and let us show that p_{k+1} is also true. By Bézout's Lemma we know that there exist integers x, y, x' and y' such that

$$1 = xa + yb \tag{3}$$

$$1 = x'a + y'b^k \tag{4}$$

Multiplying (4) by b we get that $b = x'ab + y'b^{k+1}$. Substituting this expression in (3) we get that

$$1 = xa + yb = xa + y(x'ab + y'b^{k+1}) = (x + yx'b)a + (yy')b^{k+1}$$

Applying Bézout's Lemma we obtain that $\gcd(a, b^{k+1}) = 1$. It shows that p_{k+1} is true and the result follows by the Principle Mathematical of Induction.

There is another way to prove that p_{k+1} is true assuming that p_k holds. We need to use the Primen Factorisation Theorem covered in Week 2. I was proposed this alternative solution by two different students (one in each group). Well done!

In order to prove that $\gcd(a, b^{k+1}) = 1$ we will show that there is no common prime in the prime factorisations of a and b^{k+1} . Suppose on the contrary that there exists a prime p such that $p|a$ and $p|b^{k+1}$. But then since p is prime, from $p|b^{k+1} = bb^k$ we derive that either $p|b$ or $p|b^k$. Both possibilities are impossible because $\gcd(a, b) = 1 = \gcd(a, b^k)$ (since p_1 and p_k are both true). Thus $\gcd(a, b^{k+1}) = 1$, proving that p_{k+1} holds.

2. If n and $d \neq 0$ are integers, show that there exist integers q and r such that $n = qd + r$ and $0 \leq r < |d|$.

Solution: If $d > 0$, then $|d| = d$, and this is the Division Algorithm. If $d < 0$, then $|d| = -d > 0$ and we can apply the Division Algorithm with n and $-d$ to find integers q' and r' such that

$$n = q'(-d) + r' = (-q')d + r', \text{ with } 0 \leq r' < -d = |d|.$$

Take $q = -q'$ and $r = r'$.